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United States Patent	12404648
Kind Code	B2
Date of Patent	September 02, 2025
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Trench box including modular panels and method of use thereof

Abstract

A trench box including a first plate to support a first side of a trench, a second plate to support a second side of a trench, and spreader bars extending between the first and second plates. One or both of the first and second plates is assembled from a plurality of modular panels arranged in a same plane and detachably secured to one another by a plurality of connectors and fasteners. A panel may be removed from one of the first plate and/or the second plate when an obstacle is encountered in the ground that runs across the trench. The removal of the panel creates an opening in the trench box to accommodate the obstacle without compromising the structural integrity of the trench box. Removal of the panel also does not negatively impact the trench box's ability to prevent collapse of the sides of the trench into the trench.

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Appl. No.: **17/933173**

Filed: **September 19, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240093456 A1	Mar. 21, 2024

Publication Classification

Int. Cl.: **E02D17/08** (20060101)

U.S. Cl.:

CPC **E02D17/08** (20130101);

Field of Classification Search

CPC: E02D (29/0266); E02D (29/0233); E02D (2600/30)

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Background/Summary

TECHNICAL FIELD

(1) This disclosure is directed to equipment used during excavation operations. More particularly, the disclosure is directed to a trench box that is installed in the ground during an excavation in order to hold back soil and protect workers in the trench from soil collapse. Specifically, the disclosure relates to a trench box that includes panels which may be selectively removed to work in specific regions of the sides of the trench.

BACKGROUND

(2) In order to construct buildings and install utilities it is typically necessary to excavate earth at a job site. The excavation may be accomplished by using a wide range of digging tools from backhoes to shovels which remove rocks and soil from particular areas and deposit them elsewhere. In order to lay utilities or repair utilities it may be necessary to dig a trench, i.e., a hole that is usually deeper than it is wide. Workers entering the trench to lay or repair the utilities are at risk of the sides of the trench suddenly collapsing in on them and burying them.

(3) In order to mitigate this risk it is common for a trench box to be installed in the trench as it is excavated. A trench box essentially comprises two large metal plates separated from one another by spreader bars. The ends of the trench box are open and are not covered by metal plates. Trench boxes are large and heavy and therefore have to be assembled, manipulated, installed, and removed by mechanized lifting equipment. An initial hole is dug by the digging equipment and then the trench box is installed in such a way that the metal plates are located adjacent the sides of the hole

and the spreader bars hold the metal plates a set distance apart from one another. The scoop of the digging equipment is inserted between the two plates is used to continue to remove rocks and soil. The trench box is periodically tamped down using the digging equipment and digging is continued until the desired depth of the trench is reached. In some instances additional trench boxes are installed one or both of on top of the first trench box or extend outwardly beyond one end of the first trench box. Once the desired depth and length of trench is dug, soil is backfilled between the sides of the trench and the exterior surfaces of the two metal plates. The workers can then enter the trench to perform the required work.
(4) One of the issues that is encountered while laying or repairing utilities is that in some instances pipes or conduits are arranged at angles relative to one another. Consequently, when the trench is dug and the trench box is installed the metal plates may prevent workers from accessing pipes or conduits which are oriented at an angle to the sides of the trench. In these instances the trench box has to be lifted out of the trench or moved along the length of the trench and out of the way in order for the workers to access the angled pipes or conduits. This is a time consuming and labor intensive operation and may leave workers exposed to possible soil collapse while working on the angled pipes or conduits.
SUMMARY
(5) The trench box disclosed herein is designed for use in instances where access to areas of the sides of a trench is required. The disclosed trench box permits access to these areas without the trench box having to be lifted out of the trench or be moved further along the length of the trench.
(6) A trench box and a method of use thereof are disclosed herein. The trench box includes a first plate to support a first side of a trench, a second plate to support a second side of a trench, and spreader bars extending between the first and second plates. One or both of the first and second plates is assembled from a plurality of modular panels arranged in a same plane and detachably secured to one another by a plurality of connectors and fasteners. A panel may be removed from one of the first plate and/or the second plate when an obstacle is encountered in the ground that runs across the trench. The removal of the panel creates an opening in the trench box to accommodate the obstacle without compromising the structural integrity of the trench box. Removal of the panel also does not negatively impact the trench box's ability to prevent collapse of the sides of the trench into the trench.
(7) In one aspect, an exemplary embodiment of the present disclosure may provide a trench box for installation in a trench, said trench box comprising a first plate; a second plate; and a plurality of spreader bars extending between an interior surface of the first plate and an interior surface of the second plate; wherein at least one of the first plate and the second plate is comprised of a plurality of modular panels that are detachably engaged with one another.
(8) In one embodiment, the trench box may further comprise a first connector configured to detachably secure four modular panels of the plurality of modular panels to one another. In one embodiment, the trench box may further comprise a second connector configured to detachably secure two modular panels of the plurality of modular panels to one another. In one embodiment, the plurality of modular panels in the at least one of the first plate and the second plate may be arranged in a same plane. In one embodiment at least some of the plurality of modular panels may be arranged in side-by-side abutting contact to form a row of panels. In one embodiment at least some of the plurality of modular panels may be arranged in end-to-end abutting contact to form a column of panels. In one embodiment, one of the plurality of modular panels may be detachable from the row of panels to create an opening in the row of panels or one of the plurality of the modular panels may be detachable from the column of panels to create an opening in the column of panels. The remaining panels in the row or column including the opening remain engaged with one another. In one embodiment, the at least one of the first plate and the second plate having the opening in the row of panels or the column of panels remains sufficiently strong enough to resist collapse of an adjacent side wall of the trench.

(9) In another aspect, an exemplary embodiment of the present disclosure may provide a modular panel assembly for use in assembling a side wall of a trench box; said modular panel assembly comprising a modular panel comprising a base plate having an interior surface and an exterior surface, wherein the exterior surface is adapted to be disposed adjacent a side wall of a trench and the interior surface is adapted to be disposed remote from the side wall and face into the trench; and a frame operatively engaged with one of the interior surface and the exterior surface of the base plate.

(10) In one embodiment, the modular panel assembly may further comprise a plurality of lugs extending away from the interior surface of the base plate, wherein the plurality of lugs are spaced a distance apart from one another. In one embodiment, at least one lug of the plurality of lugs may be detachably engageable with the base plate. In one embodiment, at least one lug of the plurality of lugs may be permanently secured to the base plate.

(11) In one embodiment, the modular panel assembly may further comprise a connector, wherein at least two apertures are defined in the connector spaced a distance away from one another, and wherein a first aperture of the at least two apertures may be located on the connector so as to receive one of the plurality of the lugs of the modular panel therethrough, and a second aperture of the at least two apertures may be located on the connector so as to receive one of an additional plurality of lugs of an additional and adjacent modular panel therethrough.

(12) In another aspect, and exemplary embodiment of the present disclosure may provide a method of forming a trench box comprising providing a plurality of modular panels; providing a plurality of connectors; connecting a first group of the plurality of modular panels to one another with a first group of the plurality of connectors to form a first plate of the trench box; connecting a second group of the plurality of modular panels to one another with a second group of the plurality of connectors to form a second plate of the trench box; extending a plurality of spreader bars between an interior surface of the first plate and an interior surface of the second plate.

(13) In one embodiment, the method may further comprise arranging the first group of the plurality of modular panels in a same first plane prior to connecting the first group of the plurality of connectors thereto to form the first plate of the trench box. In one embodiment, the method may further comprise arranging the second group of the plurality of modular panels in a same second plane prior to connecting the second group of the plurality of connectors thereto to form the second plate of the trench box. In one embodiment, the method may further comprise extending one or more lugs away from an interior surface of each modular panel of the plurality of modular panels; and engaging one of the plurality of connectors with a first lug of a first modular panel and with a first lug of a second modular panel, wherein the first modular panel and the second modular panel are adjacent one another in a side-by-side configuration or an end-to-end configuration. In one embodiment, extending one or more lugs away from the interior surface of each modular panel may comprise permanently securing the one or more lugs to the interior surface of an associated modular panel.

(14) In another aspect, and exemplary embodiment of the present disclosure may provide a method utilizing a trench box to prevent sides of a trench from collapsing into the trench, wherein the method comprises assembling a first plate from a plurality of modular panels and a plurality of connectors; assembling a second plate from the plurality of modular panels and the plurality of connectors; fixing a plurality of spreader bars between an interior surface of the first plate and an interior surface of the second plate to form a trench box; digging a trench in a ground to perform a work operation therein, wherein the trench extends in a first direction in the ground; determining an obstacle extends across the trench in a second direction, wherein the second direction is oriented at an angle to the first direction; removing one of the plurality of modular panels from one of the first plate and the second plate; and forming an opening in the one of the first plate and the second plate where the removed one of the plurality of modular panels was previously located to accommodate the obstacle.

(15) In one embodiment, the method may further comprise the lowering of the trench box into the trench until the first plate is adjacent a first side of the trench and the second plate is adjacent a second side of the trench; and receiving the obstacle in the opening defined in the one of the first plate and the second plate. In one embodiment, the method may further comprise supporting the first side and the second side of the trench with the trench box; and preventing collapse of the first side of the trench with the first plate and preventing collapse of the second side of the trench with the second plate, and the plurality of spreader bars.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) Sample embodiments of the present disclosure are set forth in the following description, are shown in the drawings and are particularly and distinctly pointed out and set forth in the appended claims.

(2) FIG. **1** is a top, front, left side isometric perspective view of a trench box in accordance with the present disclosure partially installed within a trench, with some of the ground around the trench box removed for clarity of illustration;

(3) FIG. **2** is a top, front, left side isometric perspective view of the trench box in accordance with the present disclosure, shown in isolation;

(4) FIG. **3** is a top, front, left side isometric perspective view of a single panel or module of the trench box, wherein the panel or module is shown in isolation;

(5) FIG. **4** is a top, front, left side isometric perspective view of a lug to be used with the panel of FIG. **3**;

(6) FIG. **5** is a top, front, left side isometric perspective view of an alternative single panel or module of the trench box, wherein the lugs are integral with or permanently and fixedly secured to the base plate of the panel;

(7) FIG. **6** is a top, front, left side isometric perspective view of a first connector for securing four panels of the trench box to one another;

(8) FIG. **7** is a top, front, left side isometric perspective view of a second connector for securing two panels of the trench box to one another;

(9) FIG. **8** is a partially exploded top, front left side isometric perspective view of the trench box of FIG. **2** with the second plate removed for clarity of illustration;

(10) FIG. **9** is an enlarged fragmentary elevation view of a top left corner region of the first plate of the trench box shown in FIG. **8**;

(11) FIG. **10**A is an enlargement of a first highlighted region of FIG. **8** showing a first connector being used to secure four panels to one another;

(12) FIG. **10**B is a cross-section of the first connector and associated panels taken along line **10**B-**10**B of FIG. **10**A;

(13) FIG. **11**A is a partially exploded top, front, left side isometric perspective view of a second connector positioned to secure two panels to one another;

(14) FIG. **11**B is an enlargement of a second highlighted region of FIG. **8** showing the second connector being used to secure two panels to one another;

(15) FIG. **12** is an exploded, fragmentary perspective view of one end of a spreader bar and a mounting bracket used to secure the spreader bar to a first plate or second plate of the trench box;

(16) FIG. **13** is a top, front, left side isometric perspective view of the trench box of FIG. **2** with a panel removed from each of the first plate and second plate; and

(17) FIG. **14** is a top, front, left side isometric perspective view of the trench box of FIG. **13** shown installed in the ground and illustrating a pipe extending through openings formed in the opposed first plate and second plate through removal of panels therefrom.

(18) Similar numbers refer to similar parts throughout the drawings.

DETAILED DESCRIPTION

(19) Referring to FIG. **1**, there is shown a trench box in accordance with the present disclosure, generally indicated at **10**. Trench box **10** is illustrated in an installed position within a trench "T" dug into the ground "G".

(20) Trench box **10** comprises a first plate **12**, a second plate **14** which are retained a set distance apart from one another by a plurality of spreader bars **16**. In accordance with an aspect of the present disclosure, each of the first plate **12** and second plate **14** comprises a plurality of individual panels **18** which are secured to one another via connector assemblies. These components will be described in greater detail hereafter. It should be noted that the number of individual panels **18** which may be included in each of the first plate **12** and second plate **14** may be varied and can be less than or greater than the number of panels **18** illustrated in the attached figures. In other words, there may be fewer than nine panels **18** in each of first plate **12** and second plate **14** or there may be nine panels as illustrated, or there may be more than nine panels in first plate **12** and second plate **14**. The first plate **12** and second plate **14** are modular in nature, being constructed from a plurality of identical modules, i.e., panels **18** and their associated connector assemblies.

(21) FIG. **3** shows a single panel **18** (or module **18**) in isolation. Panel **18** comprises a base plate **20**, a frame comprising one or more frame members **22**, and one or more brackets **24**. Base plate **20**, frame members **22**, and brackets **24** are comprised of metal. One suitable metal for fabrication of base plate **20**, frame members **22**, and brackets **24** is steel.

(22) Base plate **20** has an exterior surface **20***a*, an interior surface **20***b*, a first end **20***c*, a second end **20***d*, a first side edge **20***e*, and a second side edge **20***f*. When a plurality of modular panels **18** are assembled into a first plate **12** or second plate **14** of the trench box **10**, the exterior surfaces **20***a* of the base plates **20** of the panels **18** will be located adjacent the side walls of the trench "T" into which trench box **10** is placed. The interior surfaces **20***b* of the base plates **20** of the modular panels will face inwardly into the trench "T", i.e., away from the ground "G" which abuts the exterior surfaces **20***a* of those panels **18**. An aperture **20***g* is defined in the base plate **20** proximate each corner thereof. Each aperture **20***g* extends between exterior surface **20***a* and interior surface **20***b* of plate **20**. The purpose of apertures **20***g* will be described later herein.

(23) Panels **18** are fabricated to be of any suitable size and weight that can be readily lifted and manipulated by a construction worker but which will provide sufficient strength, when combined with the other components that form trench box **10**, to prevent collapse of the side walls of the trench "T". One suitable size for the base plate **20** of each panel **18** is a two foot by two foot square which is about 9 gauge in thickness (measured from exterior surface **20***a* of plate **20** through to interior surface **20***b* thereof). It will be understood, however, that smaller base plates or larger base plates may be utilized in the fabrication of panel **18**. Furthermore, while the base plate **20** is illustrated as being square in configuration, other shapes of base plate may be used instead. For example, the base plate could be a rectangular shape instead of being square. Additionally, while a suitable thickness for base plate **20** may be about 9 gauge sheet material, the base plate thickness may be less than a ⅛.sup.th inch or greater than a ¼.sup.th inch without departing from the scope of the present disclosure. The overall weight of panel **18** may be from about 25 lbs. up to about 150 lbs. but it will be understood that panel **18** may be of a lesser weight or of a greater weight without departing from the scope of the present disclosure.

(24) As indicated earlier herein, panel **18** includes a frame that is operatively engaged with interior surface **20***b* of base plate **20**. As illustrated, the frame comprises four frame members **22** with each frame member comprising a length of metal tubing that is welded or otherwise secured to interior surface **20***b* of base plate **20**. The frame provides structural strength and rigidity to panel **18**. It will be understood that in other embodiments, the frame members may be welded or otherwise secured to exterior surface **20***a* of each base plate **20**. In still further embodiments, some of the frame members may be welded to the exterior surface **20** and other frame members may be welded to the

interior surface **20***b*. Still further, in yet other embodiments, additional frame members may be provided and a set of frame members may be welded to the exterior surface **20***a* of the base plate **20** and a set of frame members may be welded to interior surface **20***b* of base plate **20**.
(25) As illustrated, each frame member **22** comprises a length of metal tubing which is square in cross-section. Four frame members **22** are illustrated as being welded or otherwise secured to interior surface **20***b* of base plate **20**. In particular a first frame member **22***a* is secured to interior surface proximate first end **20***c* of base plate **20**; a second frame member **22***b* is secured to interior surface **22***b* proximate second end **20***d*; a third frame member **22***c* is secured to interior surface **22***b* proximate first side edge **20***e*; and a fourth frame member **22***d* is secure to interior surface **22***b* proximate second side edge **20***f*. FIG. **3** shows that a narrow lip of base plate **20** extends outwardly beyond an outermost side of each of the frame members **22**. For example a lip **20***h* of base plate **20** extends outwardly beyond first frame member **22***a* and first end **20***c*, and a lip **20***h* of base plate **20** extends outwardly beyond fourth frame member **22***d* and second side edge **20***f*. Each of the first, second, third, and fourth frame members is of a shorter length than the respective one of the first end **20***c*, second end **20***d*, first side edge **20***e*, and second side edge **20***f* proximate which the respective frame members **22***a* through **22***d* are secured. The frame members **22***a* through **22***d* are each centered relative to the length of the associated one of the first end **20***c*, second end **20***d*, first side edge **20***e*, and second side edge **20***f*, leaving a region of the base plate **20** proximate the four corners free of an frame member **22**. Generally L-shaped brackets **24** extend between the ends of adjacent frame members **22** and are welded or otherwise secured thereto. For example a first L-shaped bracket **24***a* extends between one end **22***a′* of first frame member **24** and the adjacent end **22***d′* of fourth frame member **22***d*. The L-shaped brackets help to increase the structural strength and integrity of panel **18**.
(26) FIG. **4** shows a lug **26** which is configured to be inserted through one of the apertures **20***g* defined in base plate **20**. Lug **26** comprises a base **26***a* and a boss **26***b* extending outwardly from a surface **26***a′* of the base **26***a*. As illustrated, boss **26***b* is cylindrical in shape and is welded to surface **26***a′* of base **26***a* in such a way that the boss **26***b* is oriented at a right angle to surface **26***a′*. Boss **26***b* defines an interiorly threaded bore **26***c* that opens at a free end of boss **26***b* remote from base **26***a*.
(27) Lug **26** is configured to be inserted through one of the apertures **20***g* of base plate **20**. In particular, lug **26** is oriented such that the surface **26***a′* of the base **26***a* of lug **26** will abut the exterior surface **20***a* of base plate **20** and the cylindrical boss **26***b* will extend through the aperture **20***g* and project for a distance beyond interior surface **20***b* of base plate **20**. Cylindrical boss **20***b* extends for a short distance outwardly beyond the interior edge **24***b* of the L-shaped bracket **24** which partially bounds the associated corner region of base plate **20**.
(28) In some embodiments, lug **26** is selectively insertable through aperture **20***g* and is furthermore selectively removable from aperture **20***g*. In other embodiments, such as is illustrated in FIG. **5**, the base **26***a* of lug **26** is welded to exterior surface **20***a* of base plate **20** in such a way that lug **26** extends for a distance beyond interior surface **20***b* of base plate **20**. In particular, the cylindrical boss **26***b* extends for a short distance outwardly beyond the interior edge **24***b* of the L-shaped bracket **24** which partially bounds the respective corner region of the base plate **20**.
(29) Referring now to FIGS. **6** and **7**, two differently configured connectors may be utilized to secure the modular panels **18** to one another in order for assemble first plate **12** or second plate **14** of trench box **10** to one another. Modular panels **18** are arranged in a same plane “P**1**” (FIG. **2**) in order to ultimately be assembled together to form first plate **12** or in a same plane “P**2**” to form second plate **14**. Obviously, the modular panels **18** used in first plate **12** will be in a different plane to the modular panels used to form second plate **14**. At least some of the modular panels **18** to form first plate **12**, for example, are arranged in side-by-side abutting contact to form a row of panels **18**. Others of the modular panels **18** are arranged in end-to-end abutting contact to form a column of panels **18**. FIG. **6** shows a first connector **28** which is used to secure four adjacent individual panels

18 to one another, where those panels **18** will ultimately form part of an interior region of the first plate **12** or second plate **14**.

(30) In order to secure the four panels **18** to one another, the panels **18** are laid out in a square configuration. First panel **18***a* (FIG. **2**) and second panel **18***b* are arranged in a first row and third panel **18***c* and fourth panel **18***d* are arranged in a second row. The first and second panels **18***a* and **18***b* are arranged side-by-side, such that the second side edge **20***f* (FIG. **3**) of first panel **18***a* abuts the first side edge **20***e* of second panel **18***b*. Similarly, third and fourth panels **18***c* and **18***d* are arranged side-by-side such that the second side edge **20***f* of third panel **18***c* abuts the first side edge **20***e* of fourth panel **18***d*. The first row is placed adjacent the second row such that the second ends **20***d* of the first and second panels **18***a*, **18***b* abut the first ends **20***c* of the third and fourth panels **18***c*, **18***d*, respectively. In other words, first panel **18***a* is located in end-to-end abutting contact with third panel **18***c* to form a first column. Second panel **18***b* is located in end-to-end abutting contact with fourth panel **18***d* to form a second column. When arranged in this square configuration, the first side edges **20***e* of the first and third panels **18***a*, **18***c* are aligned with one another and the second side edges **20***f* of the second and fourth panels **18***b*, **18***d* are aligned with one another.

(31) The arrangement of the four panels **18***a*, **18***b*, **18***c*, and **18***d* also causes the corner regions at the center of the square shape to abut one another. In other words, if looking at the orientation of the panel **18** shown in FIG. **3**, the top left corner region is numbered “A”, the top right corner region is numbered “B”, the bottom right corner region is numbered “C”, and the fourth corner region, though not visible in this figure, will be numbered “D”. In the arrangement of the four panels **18***a*, **18***b*, **18***c*, **18***d*, the corner regions at the center of the square shape will comprise corner “C” of panel **18***a*, corner “A” of panel **18***b*, the corner “B” of panel **18***c*, and the corner “D” of panel **18***d*. A lug **26** is extended through each of the four apertures **20***g* in this center of the square of four panels **18***a* through **18***d*. The L-shaped brackets **24** on the four panels **18***a*, **18***b*, **18***c*, **18***d*, together, circumscribe the four abutting corner regions “C”, “A”, “B” and “D” of the panels and form a receptacle **25** (FIGS. **9** and **10**A). A lug **26** from each of the panels **18***a*, **18***b*, **18***c*, and **18***d* extends into the receptacle **25** bounded and defined by the four brackets **24**. The first connector **28** is configured to be received into receptacle **25** in order to be engaged with the four lugs **26**.

(32) First connector **28** comprises a section of tubing that is a rectangular cuboid in shape and is rectangular in cross-section. First connector **28** has a first end wall **28***a*, an opposed second end wall **28***b*, a first side wall **28***c*, and an opposed second side wall **28***d*. First and second end walls **28***a*, **28***b* and first and second side walls **28***c*, **28***d* bound and define a bore **28***e* which extends through first connector **28**.

(33) A first row of apertures **28***f*, **28***g* is defined in each of the first side wall **28***c* and second side wall **28***d*. The apertures **28***f*, **28***g* are in fluid communication with bore **28***e*. The apertures **28***f* are laterally aligned with one another and the apertures **28***g* are laterally aligned with one another. Apertures **28***f*, **28***g* are also spaced a longitudinal distance apart from one another. It should be noted that the longitudinal distance “D**1**” between the center of aperture **28***f* and the center of aperture **28***g* is substantially equal to the longitudinal distance between the centers of the lugs **26** extending from the corner region “C” of first panel **18***a* and the corner region “D” of the second panel **18***b*.

(34) A second row of apertures **28***h*, **28***j* is defined in each of the first side wall **28***c* and second side wall **28***d*. The apertures **28***h*, **28***j* are in fluid communication with bore **28***e*. The apertures **28***h* are laterally aligned with one another and the apertures **28***j* are laterally aligned with one another. Apertures **28***h*, **28***j* are also spaced a longitudinal distance apart from one another. It should be noted that the longitudinal distance “D**1**” between the center of aperture **28***h* and the center of aperture **28***j* is substantially equal to the longitudinal distance between the centers of the lugs **26** extending from the corner region “B” of the third panel **18***c* and the corner region “A” of the fourth panel **18***d*.

(35) Furthermore, as shown in FIG. **6**, the center of aperture **28***f* and the center of aperture **28***h* are

spaced a distance "D**2**" from one another. Similarly, the centers of apertures **28***g* and **28***j* are spaced the same distance "D**2**" from one another. The distance "D**2**" is substantially equal to the distance between the center of the lug **26** extending from the corner region "C" of first panel **18***a* and the corner region "B" of the third panel **18***c*. It should further be noted that the distances "D**1**" and "D**2**" are substantially equal to one another.
(36) In order to assemble first, second, third, and fourth panels, **18***a*, **18***b*, **18***c*, and **18***d*, the panels may be laid on a flat surface such that the exterior surfaces **20***a* contact the flat surface. This places all four panels **18***a* through **18***d* in a same plane. It will be understood that in other instances the panels **18***a* through **18***d* may be connected to one another when vertically oriented instead of horizontally oriented. The following description will presume the panels **18***a* through **18***d* are laid on a flat, horizontally-oriented surface.
(37) The panels are moved toward one another until the base plates **20** thereof abut one another in the configuration described earlier herein and as is shown in FIG. **9**. First connector **28** is then placed over the four lugs **26** extending outwardly from the center of the square shape formed by the four panels. In particular, the lug **26** of first panel **18***a* may be received through the aligned apertures **28***f*, the lug **26** of second panel **18***b* may be received through aligned apertures **28***g*, the lug **26** of third panel **18***c* may be received through aligned apertures **28***h*, and the lug **26** of fourth panel **18***d* may be received through aligned apertures **28***h*. Obviously, first connector **28** may be brought into contact with the lugs **26** on the four panels in a different orientation such as rotated through 180 degrees. In that instance the lug **26** of first panel **18***a* may be received through the aligned apertures **28***g* instead.
(38) Once the lugs **26** at the center of the square formation of panels **18***a* through **18***d* are received through the apertures of the first connector **28**, a washer **30** is received about boss **26***b* of each lug **26** and a fastener **32** is used to secure lug **26** to first connector **28**. The fastener may take the form of a nut that or may be a type that is externally threaded and is engaged with internally threaded bore **28***c* of each lug **26**. Fasteners **32** are tightened to a sufficient degree that the center region of the four panels **18***a* to **18***d* is secured together and will move as a unitary component.
(39) FIG. **7** shows a second connector **34** which is used to secure two adjacent individual panels **18** to one another. In particular, second connector **34** is used to secure two adjacent panels **18** to one another that will ultimately form part of a perimeter of the first plate **12** or second plate **14**. Second connector **34** is configured to be engaged with two lugs **26** provided on the two adjacent panels **18**. Second connector **34** comprises a section of tubing which is generally a rectangular cuboid in shape and has a square cross-section. Second connector **34** has a first end wall **34***a*, an opposed second end wall **34***b*, a first side wall **34***c*, and an opposed second side wall **34***d*. First and second end walls **34***a*, **34***b* and first and second side walls **34***c*, **34***d* bound and define a bore **34***e* which extends through second connector **34**.
(40) A first aperture **34***f* and a second aperture **34***g* are defined in each of the first side wall **34***c* and second side wall **34***d*, and are in fluid communication with bore **34***e*. Apertures **34***f* are laterally aligned with one another and apertures **34***g* are laterally aligned with one another. Apertures **34***f*, **34***g* are also spaced a longitudinal distance apart from one another. It should be noted that the distance "D**2**" between a center of aperture **34***f* and a center of aperture **34***g* on first side wall **34***c* is substantially equal to the distance between the centers of the lugs **26** extending from the two adjacent panels **18** with which the second connector **34** will be engaged.
(41) Referring to FIGS. **9**, **11**A, and **11**B, second fastener **34** may be utilized to secure two panels **18***e*, **18***f* to one another to form a region of the perimeter of first plate **12** of trench box **10**. Panels **18***e* and **18***f* may be laid on a flat surface such that the exterior surfaces **20***a* (FIG. **2**) of the base plates **20** thereof contact the flat surface. This places panels **18***e* and **18***f* in a same plane. It will be understood that in other instances the panels **18***e* and **18***f* may be connected to one another when vertically oriented instead of horizontally oriented.
(42) Panels **18***e* and **18***f* are moved toward one another until the base plates **20** thereof abut one

another. In particular, the second end **20***d* of panel **18***e* abuts the first end **20***c* of panel **18***f*, the first side edges **20***e* of panels **18***e* and **18***f* are aligned with one another, and the second side edges **20***f* of panels **18***e* and **18***f* are aligned with one another. In this configuration corner region "C" (FIG. **3**) of panel **18***e* is located adjacent corner region "B" of panel **18***f*. The L-shaped brackets **24** on the two adjacent panels **18***e*, **18***f* together, at least partially bound and define a receptacle **27** (FIGS. **9** and **11**A). A lug **26** from each of the panels **18***e*, **18***f* extends into the receptacle **27** and second connector **34** is configured to be received into receptacle **27** in order to be engaged with the two lugs **26**.

(43) FIG. **11**A illustrates a version of the panels where the boss **26***b* of each lug **26** is inserted through the aperture **20***g* of the associated panel corner region. It will be understood that in other versions of the panel, such as that shown in FIG. **5**, the lugs **26** are welded or otherwise fixedly secured to base plate **20**.

(44) Second connector **34** is placed over the two lugs **26** as illustrated in FIG. **11**A such that the lug **26** of panel **18***e* is received through the aligned apertures **34***f* of second connector **34** and the lug **26** of panel **18***f* is received through the aligned apertures **34***g* of second connector **34**. A washer **30** and fastener **32** is then used to lock second connector **34** to panels **18***e* and **18***f*, and thereby lock panels **18***e* and **18***f* to one another so that they will move as a unitary component.

(45) FIG. **8** shows that the plurality of panels **18** selected to form first plate **12** of trench box are secured to one another via a plurality of first connectors **28** and a plurality of second connectors **34**. Wherever the corners of four panels abut one another then a first connector **28** is utilized to secure the lugs **26** in those four corners to one another, and thereby secure the four abutting panels **18** to one another. Wherever the corners of two panels abut one another (around the perimeter of the first plate **12** that is being assembled), then a second connector is utilized to secure the lugs in those two corners to one another and thereby secure the two abutting panels **18** to one another.

(46) Referring to FIG. **8**, once first plate **12** (and second plate **14**) are formed from the desired number of panels **18** which are secured together with the appropriate first connectors **28**, second connectors **34** and fasteners **32**, a plurality of spreader bars **16** are connected to first plate **12** and second plate **14** to maintain the two plates a desired distance apart from one another. FIG. **8** shows spreader bars **16** of a fixed length being positioned for engagement with first plate **12**. However, it will be understood that adjustable length spreader bars (not shown but well known in the art) may be used instead of the fixed length spreader bars **16**.

(47) Each spreader bar **16** comprises a tube **35** and a pair of brackets **36**. Tube **35** has a first end **35***a* and a remote second end **35***b*. A mounting bracket **36** is secured to each of the first end **35***a* and second end **35***b* of tube **35**. Each mounting bracket **36** is then secured to a respective one of the first plate **12** and second plate **14**. Referring to FIG. **12** a portion of tube **35** and the mounting bracket **36** to be engaged therewith is illustrated. Tube **35** has a pair of aligned holes **35***c* (FIG. **12**) defined in each of the first end **35***a* (FIG. **8**) and second end **35***b* thereof. Only one of the two holes **35***c* is shown in FIG. **12** but it will be understood that a second hole is located diametrically opposite the hole **35***c* which is visible in this figure. Mounting bracket **36** comprises a C-shaped plate **38** having a first leg **38***a*, a second leg **38***b*, and a third leg **38***c*. First, second, and third legs **38***a*, **38***b*, **38***c* bound and define a gap **38***d* therebetween. Gap **38***d* is sized to receive the abutting frame members **22** of two adjacent panels **18** therein. For example, as shown in FIGS. **8** and **9**, mounting bracket **36** of a first of the four spreader bars **16** has a C-shaped plate **38** defining gap **38***d* sized to receive the second frame member **22***b* of first panel **18***a* and the first frame member **22***a* of third panel **18***c* therein.

(48) Mounting bracket **36** further comprises a tubular boss **40** which is welded to second leg **38***b* of C-shaped plate **38** and extends outwardly away therefrom in an opposite direction to first leg **38***a* and third leg **38***c*. Boss **40** has an exterior diameter that is complementary to an interior diameter of a bore of tube **35**. A pair of apertures **40***a* are defined through the exterior wall of boss **4** in such a way that the apertures are able to be aligned with the holes **35***c* defined in the exterior wall of tube

35 when boss **40** is received into the bore of the tube **35**. A suitable locking mechanism **42** is utilized to secure spread bar **16** and mounting bracket **36** to one another. FIG. **12** illustrates locking mechanism **42** in the form of an exteriorly-threaded bolt **42***a* and an internally-threaded nut **42***b*. When boss **40** is received within the bore of tube **35** and the holes and apertures **35***c*, **40***a* are aligned with one another, bolt **42***a* is inserted through the aligned holes **35***c*, **40***a*, and nut **42***b* is used to lock bolt **42***a* in place. In other suitable locking mechanism may be used to spreader bars **16** to the associated mounting brackets **36**.
(49) Mounting bracket **36** is secured to the panels **18** of first plate **12** (or second plate **14**) in any suitable manner. For example, mounting bracket **36** may be tack-welded, spot-welded or finally welded to the frame members **22** with which the C-shaped plate **38** is engaged. In other instances, fasteners or clamps may be utilized to secure mounting brackets **36** to panels **18** and thereby secure spreader bars **16** to first plate **12** (or second plate **14**).
(50) Once first plate **12** and second plate **14** have been assembled from the desired number of panels **18** as described herein and once the spreader bars **16** have been engaged with first plate **12** and second plate **14**, the assembled trench box **10** may be installed in the ground in the manner well known in the art. Trench box **10** may be utilized in the same manner as other known trench boxes and may subsequently be removed from the ground in substantially the same manner as is well known in the art.
(51) Trench box **10**, however, can be used in a completely unique and novel manner relative to other known trench boxes. This unique and novel manner of use is available where worker need to perform work in the trench "T" (FIG. **1**) but encounter a situation where a pipe or other obstruction intersects the trench "T" at an angle, such as at 90 degrees relative to the main trench. In these instances, one or more selected panels **18** may be removed from one or both of first plate **12** and second plate **14** as is needed. The particular panel(s) **18** to be removed from the first plate **12** (and/or second plate **14**) depends upon what obstacle is intersecting the main trench "T" at an angle. It should be noted that the number of panels **18** that are able to be removed from first plate **12** or second plate **14** depends on the rest of the plate **12**, **14** remaining sufficiently strong enough and rigid enough to prevent the side walls of the trench from being able to collapse inwardly into the trench. The number of panel(s) **18** that may be removed also depends on the overall dimensions of the first plate **12** and second plate **14**.
(52) FIGS. **13** and **14** illustrate a situation where a pipe "P" (FIG. **14**) intersects trench "T" at a right angle. Since the placement of the trench box **10** is planned thoroughly before installation of trench box **10** into the ground "G", a decision is made that it is sufficiently safe enough for one panel **18** to be removed from each of the first plate **12** and second plate **14**. Panel **18***g* is therefore removed from first plate **12**, leaving a gap **12**A defined therein. Panel **18***h* is removed from second plate **14**, leaving a gap **14**A defined therein. Panels **18***g*, **18***h* are readily removed by loosening the appropriate fastener(s) **32** and washers **30**, and sliding the lug(s) on the panel being removed from the associated first connector **28** or second connector **34**. The other panels which are going to remain as part of first plate **12** and second plate **14** remain secured to the respective first connector **28** or second connector **34**. Trench box **10** is subsequently installed in the ground "G" and pipe "P" is permitted to pass through the gaps **12**A and **14**A in trench box **10**.
(53) After work in the trench "T" is completed, trench box **10** is removed from its installed position in the ground "G" and the panels **18***g* and **18***h* are secured once again into the associated first plate **12** or second plate **14** by reversing the steps outline above. This reengagement of the panels **18***g*, **18***h* is accomplished by sliding the lugs **26** on the respective panels **18***g*, **18***h* back through the appropriate apertures in the first connector **28** or second connector **34** from which they were previously removed and then reengaging the washers **30** and fasteners **32** to once again lock the previously-removed panels **18***g*, **18***h* into the respective one of the first plate **12** and second plate **14**.
(54) It will be understood that while both the first plate **12** and second plate **14** have been described

and illustrated herein as being comprised of a plurality of modular panels which are detachably engaged with one another, in some embodiments one of the first plate and the second plate may, instead be comprised of a single sheet of metal. Utilizing a single sheet of metal for the plates of a trench box is known in the art but utilizing one plate fabricated from a plurality of detachably engaged modular panels and the other plate as a single metal sheet is not known in the art.
(55) In summary, an exemplary embodiment of the present disclosure may provide a trench box **10** comprising a first plate **12**, a second plate **14**; and a plurality of spreader bars **16** extending between an interior surface of the first plate **12** and an interior surface of the second plate **14**; wherein at least one of the first plate **12** and the second plate **14** is comprised of a plurality of modular panels **18** that are detachably engaged with one another. Trench box **10** further comprises a connector **28** configured to detachably secure four modular panels **18***a*, **18***b*, **18***c*, **18***d* of the plurality of modular panels to one another. Trench box **10** furthers comprise a second connector **34** configured to detachably secure two modular panels **18***e*, **18***f* of the plurality of modular panels **18** to one another. The plurality of modular panels **18** in first plate **12** are arranged in a first plane and the plurality of modular panels **18** in second plate **14** are arranged in a second plane. At least some of the plurality of modular panels **18** are arranged in side-by-side abutting contact to form a row of panels, such as the panels **18***a* and **18***b*. At least some of the plurality of modular panels **18** are arranged in end-to-end abutting contact to form a column of panels, such as panels **18***a* and **18***c*. One of the plurality of modular panels may be detached from a row of panels **18** to create an opening in the row of panels or one of the plurality of the modular panels **18** may be detached from a column of panels to create an opening in the column of panels. The remaining panels in the row or column including the opening remain engaged with one another. At least one of the first plate **12** and the second plate **14** having the opening in the row of panels or the column of panels remains sufficiently strong enough to resist collapse of an adjacent side wall of the trench “T”.
(56) Furthermore, the present disclosure relates to a modular panel assembly **18**, **26**, **28**, **34** for use in assembling a side wall or plate **12** or **14** of a trench box **10**. The modular panel assembly includes a modular panel **18** comprising a base plate **20** having an exterior surface **20***a* and an interior surface **20***b*. Exterior surface **20***a* is adapted to be disposed adjacent a side wall of a trench “T” and the interior surface **20***a* is adapted to be disposed remote from the side wall of the trench “T” and to face into the trench “T”. The modular panel **18** also includes a frame **22** operatively engaged with one of the interior surface **20***b* and the exterior surface **20***a* of the base plate **20**. The modular panel assembly may further comprise a plurality of lugs **26** extending away from the interior surface **20***b* of the base plate **20** wherein the plurality of lugs **26** are spaced a distance apart from one another. The modular panel assembly may further comprise a connector (such as first connector **28** and/or second connector **34**), wherein at least two apertures (such as apertures **28***f*, **28***g*, **28***h*, **28***j*, **34***f*, and **34***g*) are defined in the connector **28** or **34** spaced a distance away from one another. A first aperture of the at least two apertures, for example aperture **34***f* may be located on the connector **34** so as to receive one of the plurality of the lugs **26** of a first modular panel therethrough, such as panel **18***e*, and a second aperture, such as **34***g* of the at least two apertures may be located on the connector **34** so as to receive one of a plurality of lugs **26** of an additional and adjacent modular panel therethrough, such as panel **18***f*. The connector **28** or **34**, along with fasteners (**30** & **32**, for example) secures the first panel **18***e* and second panel **18***f* to one another in such a way that the two panels are in a same plane and are arranged in a column. Panels could, however, be secured together by the connector **28** or **34** to form a row within the ultimately assembled first plate **12** or second plate **14**.
(57) The present disclosure also relates to a method of forming a trench box **10** comprising providing a plurality of modular panels **18**; providing a plurality of connectors **28**, **34**; connecting a first group of the plurality of modular panels **18** to one another with a first group of the plurality of connectors **28**, **34** to form a first plate **12** of the trench box **10**; connecting a second group of the plurality of modular panels **18** to one another with a second group of the plurality of connectors **28**,

34 to form a second plate **14** of the trench box **10**; extending a plurality of spreader bars **16** between an interior surface of the first plate **12** and an interior surface of the second plate **14**. The disclosed method further comprises arranging the first group of the plurality of modular panels **18** in a same first plane prior to connecting the first group of the plurality of connectors **28**, **34** thereto to form the first plate **12** of the trench box **10**; and arranging the second group of the plurality of modular panels **18** in a same second plane that is different from the first plane prior to connecting the second group of the plurality of connectors **28**, **34** thereto to form the second plate **14** of the trench box **10**. When trench box **10** is ultimately assembled, the panels **18** in the first plane of the first plate **12** are arranged parallel to the panels **18** in the second plane of the second plate **14**. The disclosed method further comprises extending one or more lugs **26** away from an interior surface **20***b* of each modular panel **18** of the plurality of modular panels; and engaging one of the plurality of connectors **28**, **34** with a first lug of a first modular panel, such as **18***a*, and with a first lug of a second modular panel such as **18***b*. The first modular panel **18***a* and the second modular panel **18***b* are adjacent one another in a side-by-side configuration. In other instances, the first modular panel, such as **18***a* and the second modular panel, such as **18***c*, are arranged in an end-to-end configuration prior to being secured to one another by connectors **28**, **34**.

(58) The disclosure furthermore relates to a method utilizing a trench box **10** to prevent sides of a trench "T" dug into the ground "G" from collapsing into the trench "T". The method includes assembling a first plate **12** from a plurality of modular panels **18** and a plurality of connectors **28**, **34**; assembling a second plate **14** from the plurality of modular panels **18** and the plurality of connectors **28**, **34**; fixing a plurality of spreader bars **16** between an interior surface of the first plate **12** and an interior surface of the second plate **14** to form the trench box **10**. The method further includes digging the trench "T" in the ground "G" to perform a work operation therein, wherein the trench "T" extends in a first longitudinal direction "Y" (FIGS. **1** and **14**) in the ground "G"; determining an obstacle, such as pipe "P" that extends across the trench "T" in a second direction "X" (FIG. **14**). The second direction "X" is oriented at an angle to the first direction "Y", such as at 90° relative thereto. The method may further comprise removing one of the plurality of modular panels, such as panel **18***g*, from first plate **12** and/or removing a further one of the plurality of modular panels, such as panel **18***h* from second plate **14**; and forming an opening such as opening **12**A in first plate **12** and/or opening **14**A in second plate **14**. It will be understood that any one of the panels in either of the first plate **12** or second plate **14** may be determined to be in the way of the obstacle, pipe "P". When this determination is made the panel in the way of the obstacle is removed from the respective one of the first plate **12** and second plate **14**. It may further be determined that the obstacle requires that a panel **18** in a mirrored location or a completely different location in the other plate needs to be removed as well in order for the obstacle to be avoided by trench box **10**. So, for example, the center panel **18***g* in the bottom row may be removed from first plate **12**, but the middle panel in the middle row, **18***j*, may need to be removed from second plate **14**. In other words, an appropriate panel is removed from one or both of first plate **12** and second plate **14** to accommodate the obstacle.

(59) The disclosed method further comprises lowering the trench box **10** into the trench "T" and supporting the sides of the trench "T" with first plate **12** and second plate **14**, and thereby preventing collapse of the sides of the trench "T" into the trench itself. In the instance where a panel has been removed from first plate **12** and/or second plate **14**, the method further comprises lowering the trench box **10** with the missing panels **18** into the trench "T", supporting the sides of the trench "T" with first plate **12** and second plate **14**, and thereby preventing collapse of the sides of the trench "T" into the trench itself. In the latter instance, the disclosed method further comprises receiving the obstacle "P" in the opening **12**A or **14**A defined in the one of the first plate **12** and the second plate **14**. The method includes supporting the first side and the second side of the trench "T" with the trench box; and preventing collapse of the first side of the trench "T" with the first plate **12** (whether that plate includes all panels **18** or has one or more panels removed therefrom) and

preventing collapse of the second side of the trench “T” with the second plate (whether that plate includes all panels **18** or has one or more panels removed therefrom. The collapse is further prevented by maintaining the distance between the interior surface of first plate **12** and the interior surface of second plate **14** with spreader bars **16**. It should be noted that more than one panel **18** may only be removed from first plate **12** or second plate **14** if these plates remain of a sufficient size and strength after the removal of the panels to support the adjacent side of the trench “T” and prevent collapse thereof.

(60) The construction of the trench box **10** from modular panels **18** makes it easier for a contractor to transport the trench box **10** from a first location to a second location. The use of modular panels **18** also enables a contractor to assemble an appropriately sized trench box **10** in situ at the job site instead of having to arrange for differently sized trench boxes to be transported to the job site. The use of modular panels also enables the contractor to configure the trench box to avoid obstacles that may extend across the trench without comprising the structural integrity of the trench box. The ability to remove panels from the trench box reduces or avoids the need of a trench box to be removed from the ground to deal with the obstacle and then be replaced in the ground after the obstacle has been dealt with.

(61) Various inventive concepts may be embodied as one or more methods, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

(62) While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

(63) All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

(64) The articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” The phrase “and/or,” as used herein in the specification and in the claims (if at all), should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to

“A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc. As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.
(65) As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.
(66) It will be further understood that any connections between various components not explicitly described herein may be made through any suitable means including mechanical fasteners, or more permanent attachment means, such as welding or the like. Alternatively, where feasible and/or desirable, various components of the present disclosure may be integrally formed as a single unit.
(67) As used herein in the specification and in the claims, the term “effecting” or a phrase or claim element beginning with the term “effecting” should be understood to mean to cause something to happen or to bring something about. For example, effecting an event to occur may be caused by actions of a first party even though a second party actually performed the event or had the event occur to the second party. Stated otherwise, effecting refers to one party giving another party the tools, objects, or resources to cause an event to occur. Thus, in this example a claim element of “effecting an event to occur” would mean that a first party is giving a second party the tools or resources needed for the second party to perform the event, however the affirmative single action is the responsibility of the first party to provide the tools or resources to cause said event to occur.
(68) When a feature or element is herein referred to as being “on” another feature or element, it can be directly on the other feature or element or intervening features and/or elements may also be present. In contrast, when a feature or element is referred to as being “directly on” another feature or element, there are no intervening features or elements present. It will also be understood that, when a feature or element is referred to as being “connected”, “attached” or “coupled” to another feature or element, it can be directly connected, attached or coupled to the other feature or element or intervening features or elements may be present. In contrast, when a feature or element is referred to as being “directly connected”, “directly attached” or “directly coupled” to another feature or element, there are no intervening features or elements present. Although described or

shown with respect to one embodiment, the features and elements so described or shown can apply to other embodiments. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed “adjacent” another feature may have portions that overlap or underlie the adjacent feature.

(69) Spatially relative terms, such as “under”, “below”, “lower”, “over”, “upper”, “above”, “behind”, “in front of”, and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if a device in the figures is inverted, elements described as “under” or “beneath” other elements or features would then be oriented “over” the other elements or features. Thus, the exemplary term “under” can encompass both an orientation of over and under. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly. Similarly, the terms “upwardly”, “downwardly”, “vertical”, “horizontal”, “lateral”, “transverse”, “longitudinal”, and the like are used herein for the purpose of explanation only unless specifically indicated otherwise.

(70) Although the terms “first” and “second” may be used herein to describe various features/elements, these features/elements should not be limited by these terms, unless the context indicates otherwise. These terms may be used to distinguish one feature/element from another feature/element. Thus, a first feature/element discussed herein could be termed a second feature/element, and similarly, a second feature/element discussed herein could be termed a first feature/element without departing from the teachings of the present invention.

(71) An embodiment is an implementation or example of the present disclosure. Reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, means that a particular feature, structure, or characteristic described in connection with the embodiments is included in at least some embodiments, but not necessarily all embodiments, of the invention. The various appearances “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, are not necessarily all referring to the same embodiments.

(72) If this specification states a component, feature, structure, or characteristic “may”, “might”, or “could” be included, that particular component, feature, structure, or characteristic is not required to be included. If the specification or claim refers to “a” or “an” element, that does not mean there is only one of the element. If the specification or claims refer to “an additional” element, that does not preclude there being more than one of the additional element.

(73) As used herein in the specification and claims, including as used in the examples and unless otherwise expressly specified, all numbers may be read as if prefaced by the word “about” or “approximately,” even if the term does not expressly appear. The phrase “about” or “approximately” may be used when describing magnitude and/or position to indicate that the value and/or position described is within a reasonable expected range of values and/or positions. For example, a numeric value may have a value that is +/−0.1% of the stated value (or range of values), +/−1% of the stated value (or range of values), +/−2% of the stated value (or range of values), +/−5% of the stated value (or range of values), +/−10% of the stated value (or range of values), etc. Any numerical range recited herein is intended to include all sub-ranges subsumed therein.

(74) Additionally, the method of performing the present disclosure may occur in a sequence different than those described herein. Accordingly, no sequence of the method should be read as a limitation unless explicitly stated. It is recognizable that performing some of the steps of the method in a different order could achieve a similar result.

(75) In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed

of," and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases "consisting of" and "consisting essentially of" shall be closed or semi-closed transitional phrases, respectively.
(76) In the foregoing description, certain terms have been used for brevity, clearness, and understanding. No unnecessary limitations are to be implied therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed.
(77) Moreover, the description and illustration of various embodiments of the disclosure are examples and the disclosure is not limited to the exact details shown or described.

Claims

1. A trench box for installation in a trench, said trench box comprising: a first plate; a second plate; a plurality of spreader bars extending between an interior surface of the first plate and an interior surface of the second plate; wherein at least one of the first plate and the second plate is comprised of a plurality of modular panels that are detachably engaged with one another; and wherein a first connector is configured to detachably secure four modular panels of the plurality of modular panels to one another.
2. The trench box according to claim 1, further comprising a second connector configured to detachably secure two modular panels of the plurality of modular panels to one another.
3. The trench box according to claim 1, wherein the plurality of modular panels in the at least one of the first plate and the second plate are arranged in a same plane.
4. The trench box according to claim 1, wherein at least some of the plurality of modular panels are arranged in side-by-side abutting contact to form a row of panels or at least some of the plurality of modular panels are arranged in end-to-end abutting contact to form a column of panels.
5. The trench box according to claim 4, wherein one of the at least some of the plurality of modular panels is detachable from the row of panels to create an opening in the row of panels or one of the at least some of the plurality of modular panels is detachable from the column of panels to create an opening in the column of panels.
6. The trench box according to claim 5, wherein the at least one of the first plate and the second plate having the opening in the row of panels or the column of panels remains sufficiently strong enough to resist collapse of an adjacent side wall of the trench.
7. A modular panel assembly for use in assembling a side wall of a trench box; said modular panel assembly comprising: a panel comprising: a base plate having an interior surface and an exterior surface, wherein the exterior surface is adapted to be disposed adjacent a side wall of a trench and the interior surface is adapted to be disposed remote from the side wall and face into the trench; a frame operatively engaged with one of the interior surface and the exterior surface of the base plate; wherein the frame comprises at least two frame members; and at least one bracket extending between an end of adjacent frame members of the at least two frame members.
8. The modular panel assembly according to claim 7, further comprising a plurality of lugs extending away from the interior surface of the base plate, wherein the plurality of lugs are spaced a distance apart from one another.
9. The modular panel assembly according to claim 8, wherein at least one lug of the plurality of lugs is detachably engageable with the base plate.
10. The modular panel assembly according to claim 8, wherein at least one lug of the plurality of lugs is permanently secured to the base plate.
11. The modular panel assembly according to claim 8, further comprising a connector, wherein at least two apertures are defined in the connector spaced a distance away from one another, and wherein a first aperture of the at least two apertures is located on the connector to receive one of the plurality of lugs therethrough, and a second aperture of the at least two apertures is located on

the connector to receive one of an additional plurality of lugs of an additional and adjacent panel therethrough.

12. A method of forming a trench box comprising: providing a plurality of modular panels; providing a plurality of connectors; forming a first plate of the trench box by connecting a first group of at least four modular panels of the plurality of modular panels to one another with a first group of a first connector of the plurality of connectors and connecting a remainder of the first group of the plurality of modular panel to one another with the first group of the first connector or a first group of a second connector of the plurality of connectors; forming a second plate of the trench box by connecting a second group of the plurality of modular panels to one another with a second group of the plurality of connectors; and extending a plurality of spreader bars between an interior surface of the first plate and an interior surface of the second plate.

13. The method according to claim 12, further comprising: arranging the first group of the plurality of modular panels in a same first plane prior to connecting the first group of the plurality of connectors thereto to form the first plate of the trench box.

14. The method according to claim 12, further comprising: arranging the second group of the plurality of modular panels in a same second plane prior to connecting the second group of the plurality of connectors thereto to form the second plate of the trench box.

15. The method according to claim 12, further comprising: extending one or more lugs away from an interior surface of each modular panel of the plurality of modular panels; and engaging one of the plurality of connectors with a first lug of a first modular panel and with a first lug of a second modular panel, wherein the first modular panel and the second modular panel are adjacent one another in a side-by-side configuration or an end-to-end configuration.

16. The method according to claim 15, wherein extending one or more lugs away from the interior surface of each modular panel comprises permanently securing the one or more lugs to the interior surface of an associated modular panel.

17. The method according to claim 12, wherein forming the second plate further comprising: connecting the second group of at least four modular panels of the plurality of modular panels to one another with the second group of the first connector of the plurality of connectors and connecting a remainder of the second group of the plurality of modular panel to one another with the second group of the first connector or a second group of the second connector of the plurality of connectors.

18. A method utilizing a trench box to prevent sides of a trench from collapsing into the trench, wherein the method comprises: assembling a first plate by connecting a plurality of modular panels to one another with a plurality of connectors; assembling a second plate by connecting the plurality of modular panels to one another with the plurality of connectors; fixing a plurality of spreader bars between an interior surface of the first plate and an interior surface of the second plate to form a trench box; digging a trench in a ground to perform a work operation therein, wherein the trench extends in a first direction in the ground; determining an obstacle extends across the trench in a second direction, wherein the second direction is oriented at an angle to the first direction; removing one of the plurality of modular panels from one of the first plate and the second plate by disconnecting at least one connector of the plurality of connectors from one of the plurality of modular panels; and forming an opening in the one of the first plate and the second plate where the removed one of the plurality of modular panels was previously located to accommodate the obstacle.

19. The method according to claim 18, further comprising: lowering the trench box into the trench until the first plate is adjacent a first side of the trench and the second plate is adjacent a second side of the trench; and receiving the obstacle in the opening defined in the one of the first plate and the second plate.

20. The method according to claim 19, further comprising: supporting the first side and the second side of the trench with the trench box; and preventing collapse of the first side of the trench with

the first plate and preventing collapse of the second side of the trench with the second plate, and the plurality of spreader bars.